Evolutionary Prompt Optimization for Format-Robust Small Language Models

Advanced Computational Intelligence course project; target completion Thursday night, 29 October 2026 (Asia/Shanghai); due Friday, 30 October.

Question. Can selecting a reusable instruction for correctness across equivalent question presentations improve a frozen small language model on unseen questions and withheld formats?

Scope and contribution. Evaluate Qwen/Qwen2.5-3B-Instruct on screened, text-only, four-option ARC-Challenge questions. The incremental contribution is a controlled comparison of group-level all-presentations-correct fitness against original accuracy and augmented average accuracy. Evolutionary prompt optimization, robustness, augmentation, and racing are prior work. This project does not establish a new general optimization algorithm or reproduce full EvoPrompt, GEPA, MO-CAPO, or TARE.

Presentations and data. Five search presentations: original vertical A–D layout, inline choices, bullets, and cyclic rotations by one and two positions. An identical-input repeat is a separate control. Original labels are normalized to A–D, including ARC numeric labels; semantic identity remains the original option index. Exclude duplicate choices, position-dependent references, and possible missing media using a conservative screen. Reserve 100 train questions for the pilot, a disjoint train pool for search, official-validation questions for final prompt selection, and the screened official test split for final evaluation. Table layout, choices before the question, and reversed option order are predeclared withheld presentations, never scored during search or selection.

Methods. A: fixed manual instruction. B: random prompt proposals selected using robust fitness. C–E: common EvoPrompt-style elitist genetic search using respectively original accuracy, five-presentation average accuracy, and five-presentation all-correct fraction. Use shared initial populations, a population of four, local frozen-model mutation/crossover, uniform candidate evaluation, and three search seeds. Share a target-call budget, allowing C more candidates because it evaluates one presentation per question. Validate the top three search candidates per condition/seed using the same robust criterion, then freeze every selected prompt before final inference. No candidate is selected using test results.

Feasibility and checkpoint. Benchmark a tiny real workload before freezing the main budget. Aim to finish within one overnight allocation, with explicit tokens, proposer calls, target calls, and time. Stop if the 100-question pilot has malformed outputs, unstable identical repeats, negligible sensitivity, or effectively zero robust accuracy according to predeclared thresholds. No silent task revision. Adaptive allocation and extra models/datasets are deferred.

Evaluation. Report original accuracy, augmented average accuracy, group all-correct, invalid outputs, valid-pair semantic flips, correct-to-wrong/wrong-to-correct transitions, repeat disagreements, and separate layout/order and seen/withheld analyses. Resample whole question groups for descriptive bootstrap 95% intervals; report search-seed variation separately. Primary comparisons: E − B (evolution), D − C (augmentation exposure), E − D (fitness). Matching calls is an imperfect compute control; original-only evaluation permits more proposals. The official benchmark may overlap pretraining. A single small model and finite format set limit generalization.

Deliverables and schedule. Tuesday: implementation, integrity checks, tiny model smoke, freeze configuration. User starts the resumable overnight workflow. Wednesday: inspect saved diagnostics/results, factual automatic tables/figures. Thursday: interpret findings, inspect changed examples, complete the report and reproducibility appendix. A null or negative result remains a valid course outcome.